

ARI 410/510 Problem Set Guidelines

University of Michigan-Flint — Fall 2026

(See course Canvas page for due date)

What to do

There are several stages we will go through for these problems:

Solo: Complete the problems on your own and *submit your answers before the start of class on the due date*. Graded on completion and effort. Please answer the questions on your own, for now, based on your understanding of the course material.

Group: At the start of class on the due date, in small groups, come to a consensus about the answers. Remember who you worked with. Graded based on engagement with group members.

Class: For each question, one student will be randomly selected to present and justify the solution. The entire class may ask questions and work together as necessary to reach a solution. Graded based on participation and effort.

Final submission & Bonus: After class, make a second, updated submission with annotations regarding what changed and why. List the names of the group members you worked with and certify their engagement in the discussion. Some problem sets will have a bonus question that can be completed at this stage. Submit by the “final submission” due date. Graded based on correctness of final answers and thoughtful explanations of changes made.

What to submit to Canvas

Solo submission [first deadline]: A document with your solutions. It can be a copy of this document, or one that only contains your solutions. A PDF editor, word processor, Google Docs, or a photo of neat handwritten work are all fine, as long as your answers are clearly readable.

Final submission [second deadline]: A copy of your Solo solutions with any updates in a visually distinct way (e.g., a different color, or a comment feature), where updates include **the change and why you made it**. Also attach the completed group-member evaluation (plus answers to any bonus question(s)).

Resources

You may use resources from class or online, but the answers you submit must represent your own work and effort. **Do not use generative AI tools to directly answer the questions for you.** It is fine to ask an AI for explanations of concepts (e.g., “what is the difference between a weight and a bias term?”) or for reference information. But for a problem that asks you to derive or compute something, having an AI produce that derivation or computation counts as directly answering it. In the end, you need to be able to solve the problem on your own.

Other notes

Questions / Clarifications: ask in class or on Discord (#homework-help). **Demonstrating effort:** showing the steps you followed may be required to receive credit. **Assumptions:** if you need to make a reasonable assumption about something unclear, state it in your answer. **Levels:** most problems are shared. Where a problem forks, a [410] and a [510] marker show each track; do the one for your section. (410 students are welcome to attempt the 510 version and include it.)

Problem Set 1: Will it grow?

A community garden is trying to predict, from a couple of simple measurements, whether a newly planted seedling will **survive** its first month. They have records from earlier plantings. We will use this little dataset throughout the problem set. All plants are the same species and are planted indoors.

Plant	Sunlight (hours/day)	Water (mL/week)	Survived?
P1	3	410	No
P2	6	550	Yes
P3	7	250	Yes
P4	2	200	No
P5	4	190	No
P6	5	500	Yes

A new seedling, **Q**, has been planted with **6 hours/day** of sunlight and **400 mL/week** of water. Will it survive? That is what we will investigate.

Problem 1. k-Nearest Neighbors

We will predict Q's outcome using k-NN with ordinary (Euclidean) distance.

(a) Using the **raw** (not scaled in any way) feature values, compute the distance from Q to each of the six plants. Then classify Q using **k = 1** and using **k = 3**. Do the two values of k agree? Show your distances.

[Feel free to [check for a refresher of the formula for 2-D distance](#)]

(b) Look at what is driving those distances. Which feature is doing almost all of the work, and why? What does that mean for which feature k-NN is effectively paying attention to here?

(c) Now **standardize** each feature (subtract the mean, divide by the standard deviation) and *recompute* the $k = 1$ prediction.

To save you some calculations: please use these values:

- sunlight has mean 4.5 and standard deviation ≈ 1.7 ;
- water has mean 350 and standard deviation ≈ 144 .

(As a worked example, Q itself standardizes to sunlight $(6 - 4.5) / 1.7 \approx 0.88$ and water $(400 - 350) / 144 \approx 0.35$; standardize the six plants the same way, then recompute distances.) Does the prediction change? Explain what standardizing did and why it matters for this dataset in particular.

Problem 2. Judging a classifier

The garden built a survival model and used it to flag seedlings that are **at risk** (predicted not to survive) so a volunteer can give them extra care. On a fresh batch of 10 new seedlings, the model's flags and the actual outcomes were:

Seedling	Flagged "at risk"?	Actually died?
1	Yes	Yes
2	Yes	Yes
3	Yes	No
4	No	Yes
5	No	No
6	Yes	Yes
7	Yes	No
8	No	No
9	No	No
10	No	No

Treat **"at risk"** (flagged / actually died) as the positive class.

(a) Compute **precision**, **recall**, **F1-score**, and **accuracy**. Show your work.

(b) In this setting, which is worse: a false positive (fussing over a seedling that would have been fine) or a false negative (missing one that then dies)? Given your answer, which metric would you prioritize when tuning the model, and why?

(c) The volunteer boasts that an earlier version of the model got 100% accuracy, but it turns out they measured that accuracy on the very same plants the model had been trained on. Explain why that is a problem and what they should have done instead.

Problem 3. Logistic regression:

Instead of a hard yes/no, logistic regression predicts a probability of survival. For a plant with feature vector x , it computes $z = w \cdot x + b$ and predicts $p = \sigma(z) = 1 / (1 + e^{(-z)})$, with negative log loss $L = -[y \cdot \ln(p) + (1 - y) \cdot \ln(1 - p)]$.

Training nudges w and b “downhill” on L . The gradients (which we also used in class) are $\partial L / \partial w = (p - y) x$ and $\partial L / \partial b = (p - y)$.

Consider one plant with (standardized) features $x = (2, -1)$, meaning lots of sun and below-average water, that truly survived ($y = 1$). The current parameters are $w = (0.5, 1)$ and $b = 0$, and the learning rate is $\eta = 0.1$.

[Everyone]

(a) Compute z , then $p = \sigma(z)$, then the loss L for this plant.

(b) Compute the gradient: $\partial L / \partial w_1$, $\partial L / \partial w_2$, and $\partial L / \partial b$ using the formulas above.

(c) Take one gradient-descent step: update each weight and the bias with $w_i \leftarrow w_i - \eta \partial L / \partial w_i$ (and likewise for b). Report the new w and b .

[410 only]

(d) Using the updated w and b , compute the new z , then $p = \sigma(z)$, then the loss L for this same plant. How does this new loss compare to your answer from part (a) and why did it change in this way?

[510 only]

(d) Add regularization. We will now also penalize large weights, minimizing $L + (\lambda/2)\|w\|^2$ with $\lambda = 0.2$ (the bias is not penalized). Work out the derivative of the penalty term $(\lambda/2)\|w\|^2$ with respect to w , add it to your gradients from (b), and recompute the weight update. Compare the new weights to your answer in (c): what is the penalty doing to them?

[Need a calculus refresher? Check [here](#), in particular the “Power Rule” and “Multiplication by Constant”.
Note: we intentionally use $(\lambda/2)$ instead of just λ since this makes the derivative simpler.]

Problem 4. Support vector machines

Below are six plants in a (standardized) two-feature space; + means survived, – means did not.

Plant	Feature 1	Feature 2	Class
A	5	4	+
B	7	6	+
C	7	2	+
D	3	4	–
E	1	2	–
F	1	5	–

(a) Draw it. Plot the six points. Draw a straight line that separates the two classes. An SVM chooses the boundary to make the margin around this line as wide as possible. From your drawing, which plants sit on the margin, i.e., the *support vectors*? Name them (use the letters from the “Plant” column), and mention in a sentence what makes the SVM’s choice of boundary the “best” one.

(b) Two losses on two points. The functional *margin* of a point is $m = y \cdot z$, where $z = w \cdot x + b$ is the model's score and y is $+1$ or -1 . The hinge loss (used by SVMs) is $\max(0, 1 - m)$; the log loss (logistic regression, version that works for ± 1 labels) is $\ln(1 + e^{(-m)})$. Compute both losses for each of these two points (these are new points, not from the table above):

Point G, where $m = 2$.

Point H, where $m = 0.5$.

(c) How are these points, **G** and **H**, treated differently by the two loss functions? In general, how does this kind of difference change which set of training data points affect the decision boundary when updating parameters to minimize the loss?